

Aggregate Trends Group 2

Presentation 4 | Celena, Logan, Jake

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Potential Data Misalignment Identified

- In an ideal system, a call should end at the first instance of a row that contains a termination reason

...1	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	queue_name	agent_name	termination_reason
2084735	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Main Number Telephony EP	N/A	N/A	2025/03/17 04:53:15 PM	N/A	N/A	N/A
2084736	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	N/A	LACMain	N/A	2025/03/17 04:53:15 PM	N/A	N/A	N/A
2084737	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Main Number Telephony EP	N/A	LanguageSelectionMenu	2025/03/17 04:53:15 PM	N/A	N/A	N/A
2084738	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Main Number Telephony EP	LACMain	N/A	2025/03/17 04:53:16 PM	N/A	N/A	N/A
2084739	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Main Number Telephony EP	N/A	MainMenu	2025/03/17 04:53:26 PM	N/A	N/A	N/A
2084740	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	N/A	ClosedHoursHolidaysMenu	N/A	2025/03/17 04:54:00 PM	N/A	N/A	N/A
2084741	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Closed Hours-Holidays Menu Telephony EP	N/A	ClosedMenu	2025/03/17 04:54:00 PM	N/A	N/A	N/A
2084742	0000a8d5-cecb-46b1-82cf-b7ce07d85b24	Closed Hours-Holidays Menu Telephony EP	N/A	N/A	2025/03/17 04:54:11 PM	N/A	N/A	Customer Left

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- In an ideal system, a call should end at the first instance of a row that contains a termination reason

...	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	queue_name	agent_name	termination_reason
3166476	00057b69-8bc4-4db5-9b41-372578790b66	Main Number Telephony EP	N/A	N/A	2025/09/18 03:52:22 PM	N/A	N/A	N/A
3166477	00057b69-8bc4-4db5-9b41-372578790b66	N/A	LACMain	N/A	2025/09/18 03:52:22 PM	N/A	N/A	N/A
3166478	00057b69-8bc4-4db5-9b41-372578790b66	Main Number Telephony EP	N/A	LanguageSelectionMenu	2025/09/18 03:52:22 PM	N/A	N/A	N/A
3166479	00057b69-8bc4-4db5-9b41-372578790b66	Main Number Telephony EP	LACMain	N/A	2025/09/18 03:52:22 PM	N/A	N/A	N/A
3166481	00057b69-8bc4-4db5-9b41-372578790b66	Main Number Telephony EP	N/A	MainMenu	2025/09/18 03:52:35 PM	N/A	N/A	N/A
3166482	00057b69-8bc4-4db5-9b41-372578790b66	N/A	PreLegalMenuSeniorsMenu	N/A	2025/09/18 03:52:58 PM	N/A	N/A	N/A
3166483	00057b69-8bc4-4db5-9b41-372578790b66	Pre-Legal Menu Seniors Menu Telephony EP	N/A	SeniorsMenu	2025/09/18 03:52:58 PM	N/A	N/A	N/A
3166488	00057b69-8bc4-4db5-9b41-372578790b66	N/A	LegalMenu	N/A	2025/09/18 03:53:06 PM	N/A	N/A	N/A
3166489	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	LegalMenu1	2025/09/18 03:53:06 PM	N/A	N/A	N/A
3166497	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	LegalMenu2	2025/09/18 03:53:42 PM	N/A	N/A	N/A
3166511	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	CriminalRecordsVoicemailTransfer	2025/09/18 03:55:08 PM	N/A	N/A	N/A
3166512	00057b69-8bc4-4db5-9b41-372578790b66	N/A	N/A	N/A	2025/09/18 03:55:08 PM	Criminal Records Voicemail Transfer	N/A	N/A
3166513	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	N/A	2025/09/18 03:55:08 PM	N/A	N/A	N/A
3166514	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	N/A	2025/09/18 03:55:08 PM	Criminal Records Voicemail Transfer	CBT Agent	N/A
3166517	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	N/A	2025/09/18 03:55:26 PM	Criminal Records Voicemail Transfer	CBT Agent	N/A
3166518	00057b69-8bc4-4db5-9b41-372578790b66	Legal Menu Telephony EP	N/A	N/A	2025/09/18 03:55:46 PM	Criminal Records Voicemail Transfer	CBT Agent	Agent Left

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- However, the presence of elapsed time & multiple termination reasons following the first row with a termination reason indicates potential inconsistencies in how the call platform is logging events

...	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	queue_name	agent_name	termination_reason
2791695	001741db-98b4-409f-8646-32ed0ecf6331	N/A	Queues	N/A	2025/07/23 08:02:46 AM	N/A	N/A	N/A
2791696	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	GetLoggedInBenefitsAgents	2025/07/23 08:02:46 AM	N/A	N/A	N/A
2791697	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	IntakePreQueueMessage1	2025/07/23 08:02:46 AM	N/A	N/A	N/A
2791721	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	BenefitsQueue	2025/07/23 08:03:43 AM	N/A	N/A	N/A
2791722	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	N/A	2025/07/23 08:03:44 AM	N/A	N/A	N/A
2791723	001741db-98b4-409f-8646-32ed0ecf6331	N/A	N/A	N/A	2025/07/23 08:03:44 AM	Benefits	N/A	N/A
2791724	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	N/A	2025/07/23 08:03:44 AM	Benefits	N/A	N/A
2791725	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	PreQueueMessage2	2025/07/23 08:03:44 AM	N/A	N/A	N/A
2791727	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	PlayMOH300s	2025/07/23 08:03:56 AM	N/A	N/A	N/A
2791869	001741db-98b4-409f-8646-32ed0ecf6331	All LAC Queues Telephony EP	N/A	QueueMenu1	2025/07/23 08:08:57 AM	N/A	N/A	N/A
2791880	001741db-98b4-409f-8646-32ed0ecf6331	N/A	CourtesyCallback	N/A	2025/07/23 08:09:13 AM	N/A	N/A	N/A
2791881	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	ReadANI	2025/07/23 08:09:13 AM	N/A	N/A	N/A
2791883	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	CCB	2025/07/23 08:09:24 AM	N/A	N/A	N/A
2791884	001741db-98b4-409f-8646-32ed0ecf6331	N/A	N/A	N/A	2025/07/23 08:09:24 AM	Benefits	N/A	N/A
2791885	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	PlayCCBCConfirmation	2025/07/23 08:09:24 AM	N/A	N/A	N/A
2791888	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	DisconnectContact1	2025/07/23 08:09:33 AM	N/A	N/A	N/A
2793582	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	N/A	2025/07/23 09:13:50 AM	Benefits	Amanda Palacios	Customer Left
2793583	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	LegalServerScreenPop	2025/07/23 09:13:50 AM	N/A	N/A	N/A
2793590	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	N/A	2025/07/23 09:14:01 AM	Benefits	Amanda Palacios	Customer Left
2793862	001741db-98b4-409f-8646-32ed0ecf6331	N/A	N/A	N/A	2025/07/23 09:21:39 AM	Benefits	Amanda Palacios	N/A
2793863	001741db-98b4-409f-8646-32ed0ecf6331	Courtesy Callback Telephony EP	N/A	N/A	2025/07/23 09:21:39 AM	Benefits	N/A	Agent Left
2793904	001741db-98b4-409f-8646-32ed0ecf6331	N/A	N/A	N/A	2025/07/23 09:23:39 AM	Benefits	Amanda Palacios	N/A

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...	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	queue_name	agent_name	termination_reason
2695677	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	N/A	2025/06/10 11:58:11 AM	N/A	N/A	N/A
2695678	0004529f-0f07-421c-9d72-4baed1a8559e	N/A	Intake_Outdial	N/A	2025/06/10 11:58:11 AM	N/A	N/A	N/A
2695679	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	DisconnectContact	2025/06/10 11:58:11 AM	N/A	N/A	N/A
2695680	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	SetCallerID	2025/06/10 11:58:11 AM	N/A	N/A	N/A
2695681	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	SetCallerID	2025/06/10 11:58:11 AM	N/A	N/A	N/A
2695682	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	N/A	2025/06/10 11:58:11 AM	Intake Outdial Queue	Flora Fell	N/A
2695684	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	N/A	2025/06/10 11:58:15 AM	Intake Outdial Queue	Flora Fell	N/A
2695758	0004529f-0f07-421c-9d72-4baed1a8559e	Intake Outdial EP	N/A	N/A	2025/06/10 12:00:47 PM	Intake Outdial Queue	Flora Fell	Customer Left
2695759	0004529f-0f07-421c-9d72-4baed1a8559e	N/A	N/A	N/A	2025/06/10 12:00:47 PM	Intake Outdial Queue	Flora Fell	N/A
2695771	0004529f-0f07-421c-9d72-4baed1a8559e	N/A	N/A	N/A	2025/06/10 12:02:47 PM	Intake Outdial Queue	Flora Fell	N/A

Expected Behavior with MAX_CALLBACK_RETRY_LIMIT_REACHED

- MAX_CALLBACK_RETRY_LIMIT_REACHED and multiple CUSTOMER_UNAVAILABLE rows is expected behavior

contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	queue_name	agent_name	termination_reason
1aa8b5c3-fa9f-42af-a8c0-1d56cb	All	All	All	All	All	All	All
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:30:37 PM	Consumer	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:30:37 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:30:37 PM	Consumer	Amanda Palacios	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	LegalServerScreenPop	2025/05/23 01:30:38 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:31:04 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:31:04 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	CallbackRetry	2025/05/23 01:31:14 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:31:14 PM	Consumer	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:31:14 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:31:14 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:38:39 PM	Consumer	Darriane Sarabia	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	LegalServerScreenPop	2025/05/23 01:38:40 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:39:07 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:39:07 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	CallbackRetry	2025/05/23 01:39:17 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:39:17 PM	Consumer	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:39:17 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:39:17 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:40:49 PM	Consumer	Mona Green	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	LegalServerScreenPop	2025/05/23 01:40:50 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:41:19 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	N/A	N/A	N/A	2025/05/23 01:41:19 PM	Consumer	N/A	CUSTOMER_UNAVAILABLE
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	DisconnectCallbackContact	2025/05/23 01:41:19 PM	N/A	N/A	N/A
1aa8b5c3-fa9f-42af-a8c0-1d56cb214ac	Courtesy Callback Telephony EP	N/A	N/A	2025/05/23 01:41:19 PM	Consumer	N/A	MAX_CALLBACK_RETRY_LIMIT_REACHED

Classifying Normal vs. Irregular Sessions Logic

- Normal session:
 - Only one termination reason that is not repeated
 - `first_termination_time - last_activity_time = 0`
 - Multiple termination reasons only if the final one is `MAX_CALLBACK_RETRY_LIMIT_REACHED`, and prior ones are expected retries like `CUSTOMER_UNAVAILABLE`.
- Irregular session:
 - Multiple termination reasons
 - Termination events not in the expected order (e.g., Customer Left then Agent Left)
 - Long elapsed time after first termination event for these types

```
# Step 3: Classify sessions
session_summary <- session_summary %>%
  mutate(session_type = case_when(
    # 1) Normal if only one termination reason
    termination_count == 1 ~ "Normal",

    # 2) Normal if MAX_CALLBACK_RETRY_LIMIT_REACHED appears in that session's reasons
    map_lgl(termination_reasons, ~ "MAX_CALLBACK_RETRY_LIMIT_REACHED" %in% .x) ~ "Normal",

    # 3) Normal if first_termination_time - last_activity_time == 0
    (first_termination_time - last_activity_time) == 0 ~ "Normal",

    # 4) Everything else is Irregular
    TRUE ~ "Irregular"
  ))
```


Termination Reasons Dashboard Link

- Reveals a potential data logging inconsistency in the CARs dataset, specifically around how call session terminations are recorded
- By identifying and flagging these anomalies, we can establish a more accurate logic for logging call sessions by ensuring that sessions end at the first instance of a termination, rather than lingering on the line and potentially taking time away from getting back to other callers waiting for a callback or picking up incoming calls.

Next Steps

- Assess how these anomalies are affecting operational efficiency
- For example, assess patterns across specific agents and call types to identify whether certain conditions or individuals are driving these anomalies
- Such analyses could inform targeted process improvements or adjustments to the call platform's logging system

Thank you!
